

Question	Answer	Mark	Mark scheme	Additional guidance
22	$3 + \sqrt{2}$	P1	for start of process to find the common ratio, eg writes $\frac{13 + 9\sqrt{2}}{3 + 2\sqrt{2}}$	Award P1 for process to rationalise the denominator of $\frac{3 + 2\sqrt{2}}{13 + 9\sqrt{2}}$
		P1	for process to rationalise the denominator, eg $\frac{13 + 9\sqrt{2}}{3 + 2\sqrt{2}} \times \frac{3 - 2\sqrt{2}}{3 - 2\sqrt{2}}$	
		P1	(dep P2) for expanding terms, condone one error in expansion of numerator or denominator, eg $\frac{39 + 27\sqrt{2} - 18\sqrt{2}\sqrt{2} - 26\sqrt{2}}{9 + 6\sqrt{2} - 6\sqrt{2} - 4\sqrt{2}\sqrt{2}}$ or $\frac{39 + 27\sqrt{2} - 26\sqrt{2} - 36}{9 - 8}$ oe	
		A1	cao	A correct answer with no supportive working gets 0 marks.